A Workshop/Tutorial proposal for the

35th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2026)

August 24–28, 2026 / Kitakyushu International Conference Center, Kitakyushu, Japan

Title
Robot Fashion: Designing Wearable Identities Across Robot Morphologies (**RoboFashion**)

Format
Workshop (half day). Preferred day: **Monday, August 24, 2026** (alternatively: Thursday, August 28, 2026).

Interdisciplinary leadership. This workshop is co-led by scholars from Kent State University's **School of Fashion** (a top U.S. fashion program; fashion psychology and apparel design) and the **Advanced Telerobotics Research Lab** (HRI and social robotics), with additional industry co-organization from PlaiPin.

Main Organiser

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Co-Organisers

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Short Bio:	Industry practitioner in social robotics and embodied product design. Brings perspectives from designing modular physical AI companions and dressing robots for public-facing deployments and demonstrations.

Previous Editions

This workshop has **no prior RO-MAN editions**. It is a new offering for RO-MAN 2026, grounded in peer-reviewed HRI scholarship and ongoing practitioner momentum.

Primary grounding paper (fully cited):

Irvin Steve Cardenas, Jong Hoon Kim, Linda Ohrn-McDaniel, Krissi R. Riewe Stevenson, J.R. Campbell,

and Natalie Yeo. 2026. *Robot Fashion: The Social Psychology of Robot Clothing*. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI '26)*, March 16–19, 2026, Edinburgh, Scotland, UK. ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3757279.3788821

Reviewer note: For convenience, the full paper is attached after this proposal in the submitted PDF.

In addition, the workshop builds on community precedent in robot clothing and appearance research (e.g., design requirements for robot clothing and RO-MAN studies of clothing-driven first impressions), and on growing momentum toward public showcases, including an upcoming Robot Fashion Show planned for **September 2026** (New York City).

Statement of Objectives

Robots are increasingly deployed in public and social contexts (hospitality, retail, health, entertainment), yet attire remains an afterthought. Clothing is not cosmetic: fashion psychology shows that what an agent wears systematically shapes social cognition, including trust calibration, warmth/competence inference, role expectations, and interaction norms.

The workshop's objectives are to:

- establish robot fashion as a rigorous HRI topic with clear theoretical mechanisms (RFPM);
- bridge fashion-design expertise with HRI methods and robot engineering constraints;
- teach morphology-aware guidelines for dressing diverse robot bodies (e.g., humanoid, quadruped, tabletop, plushie/pocket companion);
- provide hands-on experience, since garment–morphology constraints and safety considerations are difficult to learn from papers alone; and
- grow a community of practice spanning robotics, fashion, design, and industry.

Intended Audience

- HRI researchers studying robot appearance, person perception, and social influence
- Robot designers/engineers working on public-facing deployments
- Fashion/textile researchers exploring non-human bodies and interaction contexts
- Industry practitioners deploying robots (hospitality, retail, events, entertainment)

Expected attendees: 25–40.

List of Speakers

Speakers/facilitators (organizers): Irvin Steve Cardenas, Jong-Hoon Kim, Linda Ohrn-McDaniel, Krissi R. Riewe Stevenson, J.R. Campbell, Natalie Yeo.

Practitioner perspectives (invited; pending confirmation): short live or recorded contributions from practitioners who dress robots in the wild (e.g., robot fight clubs, demos, deployments), potentially including Vitaly Bulatov, Nima Zeighami, and John Naulty.

Organisation. This is a workshop/tutorial hybrid (no paper submissions). The agenda alternates short, high-signal talks with an extended, hands-on making session in small groups. Each group is assigned a different robot morphology and applies RFPM and the design guidelines to produce and critique a prototype outfit.

Draft half-day schedule (09:00–13:00):

Time	Session	Speaker(s)
09:00–09:20	Welcome & why robot fashion matters (RFPM overview; workshop thesis)	Cardenas, Kim
09:20–09:50	Fashion theory for robot designers (PIDI taxonomy; dress vs. garment vs. fashion; psychology mechanisms)	Ohrn-McDaniel, Riewe Stevenson
09:50–10:20	Morphology-aware design guidelines (G1–G8) with capsule-collection show-and-tell	Campbell, Yeo
10:20–10:35	Break	—
10:35–10:55	Practitioner perspectives (lessons from dressing robots in the wild)	Invited speaker(s) / video contributions
10:55–12:15	Hands-on: dress a robot (group making session; materials provided; apply RFPM + guidelines)	All organizers (facilitators)
12:15–12:45	Show & tell + critique (fashion-semiotic + HRI critique; informal "Best Dressed Robot" vote)	All
12:45–13:00	Call to action: Robot Fashion Show (Sept. 2026, NYC) + closing discussion	Cardenas

List of Topics

Human-robot interaction; robot clothing and appearance; fashion psychology; social perception and trust calibration; morphology-aware garment design; smart textiles for robots; robot identity and branding; cultural considerations in robot attire; robot-to-robot signaling.

Plan to Stimulate Participation

- The primary draw is hands-on making: participants physically dress robots in small groups.
- Pipeline to a public showcase: participants can optionally submit designs for the September 2026 Robot Fashion Show.
- Cross-promotion through HRI/RO-MAN channels, fashion design networks, and robotics practitioner communities.
- No prior fashion experience required: guidelines, templates, and basic materials are provided.
- Invite industry partners to bring or sponsor robots and wardrobe materials.
- Post-workshop online gallery documenting outcomes and design rationales.

Statement of Inclusion, Diversity and Equity

- Materials and tools are provided at no extra cost to reduce participation barriers (enabled by Solana Foundation and PlaiPin Inc. support).
- Activities are designed for all skill levels (no prior sewing/fashion experience required).
- Multiple robot morphologies provide diverse entry points and respect different research interests.
- Cultural sensitivity in attire is explicitly taught and practiced (including discussion of norms and potential harms).
- Physical accessibility: table-height activities; seated participation possible.
- If a hybrid format is supported, talks can be livestreamed and remote participants can join critique/discussion.

- Organizer team spans disciplines (robotics, fashion, design, industry) and countries (USA, Singapore).

Workshop Organisation
Format. Short invited talks + interactive tutorial + hands-on making session with structured critique.
Submissions. No call for papers; participation is by attendance. Optional preregistration may be used for materials planning and to balance morphology group sizes.
Post-workshop. We will publish a shared design gallery (photos + short design notes) and facilitate follow-on collaboration leading into the September 2026 Robot Fashion Show.

Important Dates

- Workshop date: August 24 or August 28, 2026 (preferred: August 24).
- Pre-workshop materials (reading list + guidelines + packing list for hands-on): *TBD*(target: July 2026).
- Optional preregistration / interest form: *TBD*.

Acknowledgements
(if any) This workshop is supported by **Solana Foundation** and **PlaiPin Inc.**, with additional in-kind support anticipated from Kent State University (School of Fashion; Advanced Telerobotics Research Lab) for materials, robots, and staffing (details *TBD*).

Required In-Room Equipment

- Multimedia projector + screen.
- 4–6 demo/work tables for the hands-on session.
- Power strips/outlets for robot charging and operation.
- Standard conference Wi-Fi (no special internet requirements beyond typical connectivity).

Robots (organizer-provided; subject to travel feasibility): humanoid (e.g., Pepper or Unitree G1), quadruped (Unitree Go2), tabletop (KAMI), plushie/pocket companion platform (PlaiPin).
Craft supplies (organizer-provided): fabric swatches, scissors, Velcro, elastic, snaps, pins, tape, and basic sewing kits.

Robot Fashion: The Social Psychology of Robot Clothing

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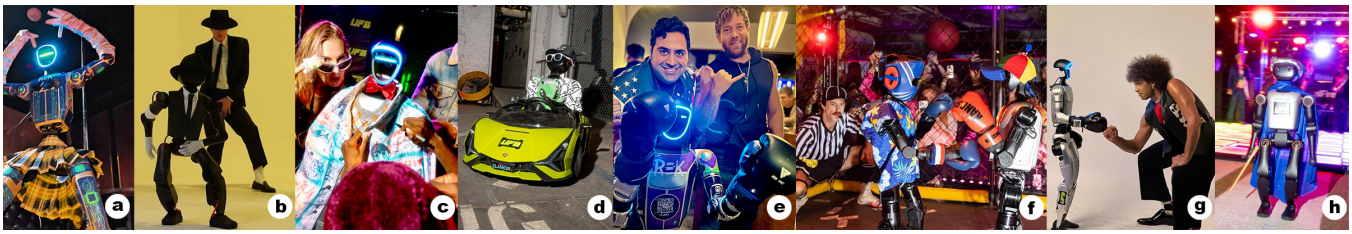

Figure 1: Robots dressed in practice: (A–D) entertainment and transgressive costume; (E–H) fight-club attire (Sec. 4).

Abstract

Most robots today are naked, and nobody seems to mind! We argue they should. Robot attire is not cosmetic: clothing systematically shapes social cognition, cueing role expectations, warmth and competence inferences, touch norms, and trust. We introduce the *Robot Fashion Psychology Model (RFPM)*, a mid-level theory linking what robots wear to how people interact with them, moderated by morphology and context. Grounded in fashion-led research-through-design that triangulates in-the-wild exemplars, a capsule collection spanning humanoids to drones, and reflexive critique – we translate RFPM into morphology aware garment guidelines and ethical guardrails against "trust-washing" and capability deception. We then ask: What happens when robots dress for each other? When fashion becomes visible only to machines? Robot wardrobes are coming. With them come new forms of designed influence. We offer foundations for designing them with intention rather than improvisation, while norms are still forming.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; **HCI design and evaluation methods**; • **Computer systems organization** → *External interfaces for robotics*; *Robotics*; • **Applied computing** → *Media arts*; *Psychology*.

Keywords

Human-robot interaction, robot clothing, dress and fashion theory, Robot Fashion Psychology Model (RFPM), social signaling

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1 Introduction

Most robots today are built with exposed metal, plastic, or composite exteriors – essentially *unclothed* machines. At first glance, this appears unproblematic: unlike humans, robots feel no modesty, and society has not expected them to wear pants or dresses. However, research suggests that robot clothing presents a powerful, *underutilized opportunity* in robotics and design [26]. Clothing can help robots "become better robots" by enabling them to function in more contexts and communicate more effectively [26]. Traditionally, robot appearance has been considered mainly in terms of chassis form factor or anthropomorphic shape, but we argue that *garments and adornments* can play equally pivotal roles. This paper introduces *robot fashion* as a concept that extends beyond utilitarian coverings or cosplay-like decoration, proposing that what robots wear will influence not only human perceptions but potentially robots' perceptions of each other in multi-robot ecologies.

To establish psychological foundations for this emerging domain, we propose the **Robot Fashion Psychology Model (RFPM)** (Sec. 4.1; Fig. 2), a mid-level theory explaining how robot attire shapes social perception and interaction.

1.1 Key terms: Garment, Dress, and Fashion

We adopt distinctions from fashion studies: **garment/clothing** denotes textile or shell objects on bodies [38, 42]; **dress** denotes the practice of covering, protecting, and styling bodies [21, 51]; and **fashion** denotes the sociocultural system of codes and institutions that classify what is "on-trend" [4, 23, 38, 48, 49]. This vocabulary

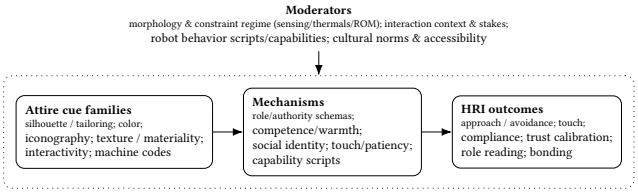

Figure 2: RFPM pipeline. Attire cues activate psychological mechanisms that shape HRI outcomes; morphology, context, behavior, and culture moderate these pathways. We operationalize it through the PIDI taxonomy (Tab. 2), the mechanism map (Tab. 1), and garment guidelines (Tab. 3).

clarifies how PIDI functions (Table 2) describe what robot attire *does*, while RFPM (Fig. 2; Table 1) explains *how* attire cues produce psychological effects, motivating design guidance (Table 3).

Our thesis is that *robot fashion will increasingly matter for robot-to-robot perception and social integration*, not merely for pleasing human observers – with clothing becoming a multi-level communication medium (e.g. drones wearing "jerseys" for fleet coordination while signaling affiliation to passersby). Early practice signals support this view: robots are already dressed for public events, community contexts, and deployments, where even minimal accessories shift impressions and interaction norms [36]. HRI design principles for robot garments [26], smart textile advances (thermochromic threads, capacitive sensing) [19, 39], and attire-driven stereotype research all contribute to an emerging foundation.

In this paper, we blend research-through-design with speculative design to address: how fashion applies across robot morphologies, what signals robot clothes send to humans and other robots, how functional constraints (thermals, sensing) interact with expression, who owns a robot's appearance, and whether robots might develop fashion sensibilities in domains only they perceive.

Our contribution is threefold. *First*, we introduce RFPM (Fig. 2) as a psychological foundation for robot fashion and summarize key cue→mechanism→outcome pathways in a mechanism map (Table 1). *Second*, we present a morphology-aware framework that links material constraints to social functions via PIDI (Table 2) and translate RFPM into morphology-aware garment guidelines (Table 3) grounded in a capsule collection and in-the-wild exemplars. *Third*, we extend these grounded foundations with speculative scenarios and implication mechanisms (e.g. accountable wardrobe autonomy) to stress-test RFPM, surface ethical tensions, and articulate testable research directions for future robot wardrobes.

2 Related Work

2.1 Robot Appearance and Clothing in HRI

HRI has long examined how a robot's *appearance* modulates attitudes and behavior toward it (e.g. anthropomorphic vs. machine-like form, facial displays, material finish). *Clothing*, however, has only recently been foregrounded as a first-class design variable. Prior work argues that robot garments should not copy human fashion heuristics but be designed for robots' distinct thermals, kinematics, sensing, and maintenance constraints, outlining functional requirements and patterning strategies that adapt robots to new contexts, protect hardware, and even serve as interaction surfaces [26]. Follow-on community work has begun to catalog *why* practitioners dress robots – from signaling role and affiliation, to softening eeriness, to durability and serviceability, and *how* (skins, covers, costumes, and modular add-ons) [24].

Appearance cues via clothing also shape *first impressions* and social stereotyping: outfitting a humanoid in a dress vs. a suit shifts perceived gender, human-likeness, occupational fit, and competence/warmth judgments [36]. Parallel social-psychology syntheses argue that dress is a fundamental component of person perception used to infer group membership, status, intent, and taste [35]. In developmental HRI, letting children co-create apparel for a NAO increased self-disclosure and perceived rapport, suggesting clothing can scaffold bonding and intentionality attributions ("it wore what I made") [27]. Beyond lab studies, community showcases and exhibitions have surfaced design practices (e.g. seam allowances around joints, venting paths, sensor windows) and dramaturgical aims (softening industrial affordances, imbuing role) [18, 24, 25]. A recurring principle is morphology-specific design: ill-fitting human garments impede motion or occlude sensors; patterns must be drafted for robot body plans (humanoid, quadruped, aerial, continuum) rather than ported from human blocks [26].

2.2 Fashion Theory: Resources and Gaps for Robot Attire

Fashion studies and social psychology provide powerful tools for understanding how clothing shapes perception and interaction – but these frameworks were built for human wearers and human audiences. We draw on this literature while identifying three critical gaps that robot fashion must address.

What Fashion Theory Offers. Fashion psychology establishes that dress is not trivial: it systematically shapes identity, social perception, and well-being, despite cultural tendencies to dismiss it as superficial [34, 40]. Dress functions as communicative media, cueing inferences about group membership, status, intent, and taste [35]. Within the stereotype content model, silhouettes and styling reliably shift competence and warmth judgments [22, 36]. Semiotic accounts show that garments become meaningful through culturally elaborated codes (e.g. palettes, silhouettes, insignia) circulated in discourse [4]. Dramaturgical perspectives frame dress as part of a performed "front" that coordinates expectations about roles and competencies [28]. And embodiment theorists emphasize that dress achieves fullness only on and through bodies, shaping movement, posture, and social legibility [21]. These accounts converge: clothing recruits cognitive shortcuts that shape how we read and interact with dressed bodies.

Gap 1: Fashion Theory Assumes Human Wearers. Symbolic interactionist accounts treat fashion as a "social object" whose meanings are *negotiated* through interaction and stabilized via "collective selection" [2, 8]. But robots do not negotiate their wardrobes – designers, operators, or institutional policies do. Sociological accounts frame fashion as a system of production (designers, journalists, brands, retail) that elevates certain garments to "fashion" status [38]; no such institutional apparatus exists for robots. Simmel's classic tension between *affiliation* and *differentiation* assumes wearers who seek belonging and distinction [49], motivations robots lack.

The question for robot fashion is not what the robot wants to express, but *who authors the robot's appearance* and *whose interests it serves*. Actor-network theory positions robots as actors that "do things and enable us to do other things" [20]; their dressed appearance participates in distributed agency, yet unlike human actors, robots cannot author their own presentation. This raises questions of the "gaze": who dresses the robot, for whom, and through what interpretive lens [44] – reminding us that robot dress carries social implications that can reinforce or challenge existing constructs. This gap motivates our attention to ownership, governance, and the principal-agent structure of wardrobes (Sec. 4.5).

Gap 2: Fashion theory assumes wearer agency. Goffman's dramaturgy assumes the *performer* manages their own impression; the wearer chooses a "front" to support a desired reading [28]. For robots, the performer (robot) and costume designer (human) are decoupled. This creates a *principal-agent problem*: appearance may serve operator or brand interests rather than interaction quality. Relatedly, *enclothed cognition* – the joint effect of a garment's symbolic meaning and the embodied experience of wearing it – has shown effects on human wearers' cognition, though recent meta-analyses reveal fragile replication [1, 37]. For robots, the relevant construct is not the robot's internal state but humans' *enclothed attributions*: clothing shifts *observers'* expectations about competence, intent, and permissible interaction. When attire implies capabilities a robot lacks, expectancy violations produce "trust-washing" and over-trust – making *capability-honest* styling a design imperative (G5). Clothing also participates in identity ownership: personalized garments can become part of the owner's *extended self* [5, 47], which helps explain bonding effects in co-creation settings [27] but raises questions about who "owns" a robot's look.

Gap 3: Fashion Theory Assumes Shared Human Codes. Fashion cycles operate as status games of distinction and diffusion within human social hierarchies [41]; nonconformity can signal status in certain contexts [6]. But robots exist in human–robot dyads where the "reader" (human) and "wearer" (robot) are ontologically different, and increasingly in multi-robot ecologies where robots may signal to *each other*. Traditional dress theory has no account of machine-legible codes (IR/UV patterns, fiducials) visible to robots but not humans. This extension, *robot-facing fashion*, is novel: it enables coordination benefits but introduces auditability concerns when humans cannot inspect what is being signaled. Emerging digital fashion and product passports further complicate authenticity via verifiable provenance [3, 10], suggesting wardrobe histories and governance structures that could extend to robots.

From Gaps to Mechanisms. Despite these gaps, the core insight of fashion psychology holds: attire cues recruit cognitive shortcuts that shape interaction. We distill recurring *mechanism families* into the foundations of the *Robot Fashion Psychology Model (RFPM)* (Sec. 4.1), each addressing how fashion theory's resources must be adapted for robot contexts:

- **Role schemas & authority heuristics:** uniforms and "official" cues accelerate role inference and increase compliance – but also raise over-trust risk when the authoring agent (operator) is decoupled from the performing agent (robot) [7, 22, 28].
- **Competence / warmth inferences:** silhouettes and styling shift perceived competence and warmth; because robots lack self presentation goals, designers bear responsibility for "capability-honest" appearance [22, 36].
- **Social identity & categorization:** iconography, colors, and stylistic codes signal affiliation. But, meanings are culturally contingent and can exclude outsiders [8, 35].
- **Touch norms & moral patiency:** materiality and texture shape what feels "touchable" and how much moral concern observers extend; soft exteriors may invite touch but risk infantilization or trust-washing in high-stakes domains.
- **Capability scripts & expectancy management:** dress cues what people think a robot *should do*; expectancy violations occur when appearance implies a script the system cannot execute [28].
- **Robot-facing codes:** unlike traditional dress, robots can wear machine-legible cues that support coordination while raising transparency concerns – a mechanism family with no precedent in fashion theory.

This distillation explains why robot fashion is not merely aesthetic: attire cues reliably activate psychological pathways that shape interaction, and these pathways must be designed under morphology, safety, and governance constraints. Our PIDI framing (Table 2) separates material functions (Protection/Performance) from coded social functions (Identification/Communication; Decoration/Expression; Integration/Culture), clarifying why safety overlays and role honesty must remain legible regardless of the decorative layer.

2.3 Materials Science and Smart Textiles

Smart textiles afford dynamic, interactive exteriors that merge function and expression. Thermochromic threads and light-emitting garments enable color shifts and affective displays [19, 50]; e-textile stacks integrate sensing, actuation, and communication, supporting tactile interaction without rigid skins [39]. Wearable telepresence systems demonstrate how garment-like interfaces can couple human operators to humanoid robots with sensory feedback [15]. Beyond expression, textiles provide insulation, protection, and ventilation while preserving sensor fields and joint ranges [26]. Modular and camouflaging exteriors enable task-specific reconfiguration and contextual signaling. Collectively, these materials position garments as multifunctional interfaces rather than inert decoration.

3 Approach & Evidence

We use a *fashion-led, research-through-design* approach to examine robot fashion as a material, technical, and social phenomenon. Rather than testing a single hypothesis, we aim for *traceable, generative* insights by triangulating: (i) grounding in scholarship and situated practice, (ii) artifact-centered prototyping across robot morphologies, and (iii) speculative narrative prototyping to surface implications, risks, and research questions. This process yields synthesis artifacts that structure the paper's foundations: RFPM (Fig. 2) and its mechanism map (Table 1), alongside morphology-aware garment guidelines (Table 3).

Positionality and Expertise. Our author team spans fashion faculty (garment construction, patternmaking/knitwear, and fashion theory) and robotics/HRI researchers (robot design, deployment, and interaction). We draw on fashion-semiotic analysis alongside HRI perspectives on appearance, signaling, and social perception;

Table 1: RFPM mechanism map: attire cue families activate psychological mechanisms that shape HRI outcomes. Mechanism families are grounded in prior work across dress perception, impression management, robot clothing effects, compliance/authority effects, and mind perception/moral patiency. "Countermeasures" point guidelines (G1 - G8, Table 3).

Attire cue family	Psychological mechanism	Expected HRI outcomes	Primary risks & countermeasures
Uniform/livery cues: (hi-vis vests, insignia, ID badges, "official" palettes)	Role-schema activation; authority heuristics; fast competence/legitimacy inference.[7, 11, 12, 17, 22, 28, 35, 43]	Increased compliance/deference; clearer role interpretation; reduced public uncertainty.	*Over-trust / credential confusion* when attire implies capabilities the robot lacks. **Countermeasures:** G4, G5, G7.
Silhouette/tailoring: (sharp vs soft lines; "professional" vs playful archetypes)	Competence/status inferences; "capability scripts"; expectancy shaping.[1, 22, 28, 36, 37]	Shifts in perceived professionalism/authority; altered expectations; stronger error reactions under "high-competence" looks.	*Expectancy violations* if behavior mismatches implied role; gendered/normative readings. **Countermeasures:** G5, G7.
Materiality/texture: (plush/knit/fleece vs hard-shell "tactical" materials)	Warmth/patiency attributions; touch norms; pet/worker schema activation.[22, 24, 29–32, 53]	Increased approachability/touch for soft exteriors; reluctance to mistreat; "pet" vs "working" scripts.	*Trust-washing* in high-stakes domains; sensory overload; unwanted touch. **Countermeasures:** G4, G7, G1.
Iconography/affiliation: (team marks, faction colors, cultural symbols, novelty costumes)	Social identity/categorization; subculture signaling; persona construction.[2, 4, 8, 35, 41]	Faster bonding/naming; spectator legibility; clearer affiliation boundaries.	*Exclusion / bias* via insider codes; misread symbolism; appropriation. **Countermeasures:** G7, G5.
Interactivity in attire: (light, thermochromic, touch sensing, slow displays)	Feedback loops: attire as readable state layer supporting emotion attribution, turn-taking, touch conventions.[19, 39, 50]	Increased state/intent legibility; smoother pacing; richer nonverbal feedback; new touch etiquette.	*Signal clutter*; misinterpretation; added failure modes. **Countermeasures:** G6, G3, G2.
Robot-facing codes: (IR/UV patterns, fiducials, optical codes)	Identification/coordination reducing human-visible clutter; "machine-vision-only fashion" as ecology layer.[18, 50]	Improved fleet coordination/recognition; reduced robot-robot ambiguity; privacy benefits for humans.	*Opacity / auditability* (humans cannot inspect signals). **Countermeasures:** G4; bounded policy enforcement (see supplementary).

this interdisciplinary position shapes what we foreground (material constraints and social codes) and how we interpret evidence (as illustrative themes and design implications rather than population-level causal effects).

Grounding (literature + situated exemplars). We reviewed HRI first impressions and social cognition [35, 36], functional and patterning requirements for robot garments [24, 26], and interactive/smart textile systems [19, 39, 50]. In parallel, we documented real-world instances where robots are dressed in public and professional contexts, treating these as *situated exemplars* revealing expectations, norms, and constraints (illustrative rather than representative).

Practitioner Perspectives. We gathered practitioner perspectives through informal conversations with people who dress robots (fight club operators, robotics company staff), using shared images to prompt discussion of design motivation, constraints, and audience reactions. These perspectives inform Sec. 4; our discussion guide is in supplementary materials.

Capsule Collection (artifact iteration). We fabricated a morphology-diverse capsule collection (humanoid, quadruped, aerial, tabletop) using digital patterning, sewing, and e-textile techniques. Iterations targeted do-no-harm constraints, signaling legibility, modularity, and cultural sensitivity [24, 26, 36].

Speculative Scenario Development. To extend beyond near-term constraints, we authored narrative prototypes and low-fi enactments (robot–robot etiquette, wardrobe autonomy and garment passports, and machine-vision fashion via IR/UV or optical codes). Scenarios were critiqued against HRI ethics (deception/transparency), social-psychological plausibility, and technical feasibility (materials, power, maintenance).

Analysis Strategy and Limitations. We employed abductive, reflexive analysis common to research-through-design: artifacts, documentation, and scenarios functioned as boundary objects to surface constructs and tensions. We began from the mechanism families distilled from fashion psychology and HRI in Sec. 2 and iteratively refined them into RFPM pathways (Fig. 2; Table 1) through triangulation with prototype iterations, practitioner feedback, and observed failure modes. We then mapped RFPM pathways to morphology-aware guideline countermeasures (Table 3). We report themes and design guidelines rather than hypothesis tests; practice documentation is event- and visibility-biased and should be read as illustrative.

4 Themes and Design Implications

We synthesize themes from our evidence sources (Sec. 3): literature, situated exemplars of robots being dressed in practice (including photo documentation and practitioner conversations), artifact iteration through a morphology-diverse capsule collection, reflexive critique sessions, and speculative scenario work. These themes are reported as *illustrative patterns and design implications* (not population-level prevalence claims or causal effects), and they motivate the morphology-aware guidelines summarized in Table 3.

Analytic Lens for "in-the-wild" Exemplars. To interpret fight-club and industry exemplars (Fig. 1), we combine a fashion-semiotic lens (e.g. *uniform/role*, *costume/character*, *affiliation / subculture*, *status/taste*, *gender coding/normativity*) with an HRI lens (e.g. *warmth / competence impressions*, *approach/avoidance and touch norms*, *legibility/safety*, and *trust calibration*). We use image-based prompts and informal conversations (see supplementary materials) to connect visual choices to intent, constraints, and observed audience reactions.

4.1 Robot Fashion Psychology Model (RFPM)

To make robot fashion legible as a first-class HRI design variable, we propose the **Robot Fashion Psychology Model (RFPM)**: a mid-level theory that links *attire cues* to *psychological mechanisms* and *interaction outcomes*, with *morphology, context, and behavior* acting as moderators. RFPM bridges fashion psychology (dress as meaning-making and identity performance) with social cognition in HRI (appearance-based impression formation, role inference, and trust calibration). Figure 2 summarizes RFPM as a pipeline; Table 1 instantiates common cue→mechanism→outcome pathways and their design countermeasures.

Table 2: Robot fashion taxonomy across morphologies. Functions encoded as PIDI: Protection/Performance (P), Identification/Communication (I), Decoration/Expression (D), Integration/Culture (In). H/R indicates human vs robot-facing signals.

Morphology	Garment modality (examples)	Primary functions (PIDI)	Key patterning & engineering constraints	Signaling channels (H / R)
Humanoid	tailored jackets, trousers, dresses, vests	In, I, D	shoulder/elbow gussets; back vents for thermals; camera windows; preserve ROM and FOV	**H:** palette, silhouette, uniform cues, ID badge. **R:** chest/shoulder fiducials; IR/UV marks; status LEDs.
Quadruped	fleece wraps; modular shells; harnesses	P, D, In	leg clearance; spine access; righting; segmented cuffs; reflective piping; fast don/doff	**H:** fleece (pet) vs hard-shell (K9) livery. **R:** side-panel fiducials; tail/withers LEDs.
Aerial (drones)	ripstop skins; ultralight shells; decals	I, P	rotor-wash cutouts; weight/CG budget; airflow paths; quick-swap mounts	**H:** high-vis palettes; top/bottom livery for altitude. **R:** top-plane patterns; IR landing codes.
Tabletop / Wheeled	plush covers; capes; decals / hoods	D, I, In	snag prevention; cooling vents; preserve FOV and mics; removable liner	**H:** texture for touch; approachable colors; signage. **R:** collar/brow fiducials; low-power status lights.
Continuum / Biomorphic	knit "second skins"; segmented sleeves	P, D	segment allowances; distributed stretch; avoid seam abrasion; embedded sensing	**H:** calming textures; minimal clutter. **R:** along-body IR/UV codes; distributed touch sensing.

Model Components. RFPM comprises four parts: (1) **Attire cues** (silhouette, colorways, iconography, texture, uniform markers, interactivity, and machine-legible codes); (2) **Mechanisms** that fashion and social psychology suggest are activated by dress (e.g. role-schema activation and impression management; competence / warmth inferences; social identity and categorization; touch norms and moral patiency; "capability scripts" that people project onto dressed bodies); (3) **HRI outcomes** (approach/avoidance, touch willingness, compliance, perceived authority, trust calibration, and expectancy violations); and (4) **Moderators** (morphology and constraint regimes, the robot's actual behavior scripts and domain stakes, and cultural context).

Mechanism Map. Table 1 summarizes cue → mechanism → outcome pathways, their ethical risks, and countermeasures (G1 – G8, Table 3) that constrain or leverage these effects.

From Model to Design Guidance. RFPM is operationalized via morphology-aware garment guidelines (Table 3). In brief: G1–G3 and G8 constrain *feasibility* (ensuring cues can be expressed without harming sensing/thermals/kinematics or maintenance), while G4 – G7 constrain *meaning* (ensuring cues are legible, capability-honest, culturally sensitive, and ethically bounded). The remaining subsections show how moderators shape what is wearable, how mechanisms appear in practice, and how this motivates both design guardrails and testable hypotheses.

4.2 Moderators: Dressing Across Robot Morphologies

In RFPM (Fig. 2), *morphology is a moderator*: it governs what can be worn without harming sensing, kinematics, and thermals, and it shapes which cue families are available (e.g. tailoring cues are feasible on humanoids but largely infeasible on drones). Robots span humanoids, animaloids (e.g. quadrupeds), wheeled platforms, aerial drones, and continuum/biomorphic forms. A recurring constraint is that garment logic must be *morphology-specific*: patterns, seam placement, attachment, and allowances must be drafted for robot kinematics, thermals, and sensor fields rather than ported from human blocks [24, 26]. This is also where PIDI becomes practical: the same function (e.g. *Identification*) must be implemented through different cue families depending on morphology.

Humanoids. Because humanoids share limb/topology analogies with people, tailoring and silhouette cues (RFPM: *silhouette/tailoring*) can be viable with adaptations: stretch gussets at shoulders/elbows, back vents for heat, and windows for cameras or status LEDs. In our prototypes and critique sessions, business-casual attire was often interpreted as a "friendly office assistant," while suiting and sharper silhouettes were interpreted as more formal or authoritative. Clothing also shifted gendered attributions and role expectations, consistent with prior findings that robot clothing modulates occupational fit and gendered reading [36]. *However, many humanoids remain difficult to dress* due to harnessing, access-panel requirements, thermal constraints, and protruding hardware; our examples illustrate feasibility in specific configurations rather than a universal assumption (G1–G3). *"I've never seen a reaction like that to a robot demo in my 10 years of robot career. Drones, rovers, industrial arms don't cause as much emotional reaction. But with humanoids, people hug them; especially older ladies kiss them"* – Vitaly Bulatov, Ultimate Fighting Bots [52].

Quadrupeds. Human clothes fit poorly; more apt are dog coats, horse blankets, harnesses, or modular shells. Quadrupeds foreground materiality and texture cues (RFPM: *materiality/texture*) that rapidly trigger pet/worker schemas: soft textiles elicited pet-like treatment (e.g. stroking, "baby-talk"), while hard-shell or "tactical" covers read as a working K9. Patterns must preserve leg clearance, spine access, and righting (G1–G3).

Aerial Robots (drones). "Clothing" manifests as ultralight skins and shells with prop-wash cutouts. Drones privilege livery and visibility cues (RFPM: *uniform/livery*; *visibility cues*) and robot-facing codes (RFPM: *machine-legible codes*) under strict weight/airflow constraints. High-visibility palettes communicate affiliation/safety; computer-vision-recognizable patterns support identification.

Continuum and Wheeled Robots. Sleeves and second-skins (knit meshes with segment allowances) can add tactile qualities and sensing; wheeled kiosks admit decals, hats, capes, and projected skins. In critique sessions, even minimal accessories (e.g. a bow tie) were sufficient to prompt character attributions, demonstrating that small iconography cues can produce outsized meaning shifts when morphology leaves observers uncertain.

Situated Exemplar (fight club; morphology pressure). In fight-club settings, teams often dress robots under constraints that intensify morphology concerns: rapid repair, impact tolerance, and audience legibility under motion and lighting (Fig. 1). Practitioners described using *modular* parts (e.g. quick-release panels, magnets, strap-on badges) because garments must be removed between bouts and cannot obstruct maintenance access – a lifecycle reality that directly motivates serviceability and don/doff guidance (G3, G8). *"We came up with a strongman suit. Full tight, covering all over. The robot overheated mid-event and stopped responding. We had to cut the leggings off and swap to another robot. Something on the board got damaged from overheating"* – Vitaly Bulatov, Ultimate Fighting Bots [52].

Takeaway. Morphology-tailored attire supports different primary goals by enabling different cue families: *integration* via silhouette/tailoring on

humanoids, *characterization* via texture/materiality on animaloids, and *identification/protection* via livery and machine-legible codes on drones and service platforms. Across contexts, the same fashion function manifests through different material tactics because morphology determines what can be covered, how it attaches, and what must remain visible or vented.

4.3 Mechanisms in Action: Impression Formation, Identity, and Trust

Dress is a fundamental channel in person perception – supporting inferences about group, status, intent, and taste [35]. Across artifacts, in-the-wild exemplars, and critique sessions, robot attire shaped first impressions, role attributions, and touch norms, aligning with HRI results that clothing modulates human-likeness and occupational fit [36]. In RFPM terms (Fig. 2; Table 1), attire cues activate role schemas, competence/warmth inferences, and capability scripts that set expectations for interaction.

First Impressions (competence/warmth). Undressed robots were read as "robotic" or "bare"; casual outfits were often read as more approachable, and tailored jackets as more professional. These shifts mirror attire-driven expertise/likability effects reported in [36] (RFPM: *silhouette/tailoring* → competence scripts; *texture/materiality* → warmth/touch norms).

Role Schemas and Compliance. Uniforms, colorways, and iconography (e.g. municipal vests) function as legible codes for role/affiliation. Consistent livery across fleets can support wayfinding and compliance (e.g. yielding right-of-way), but also risks over-signaling authority (RFPM: *uniform/livery* → authority heuristics → compliance). This motivates explicit disclosure and capability-honest design (G4–G5).

Personalization and Bonding. Small accessories (scarves, pins) prompted naming and bonding, echoing child co-creation work where dressing robots scaffolds rapport and self-disclosure [27] (RFPM: *iconography/affiliation* → social identity → bonding). Figure 3a shows how minimal cue changes can re-cast persona while leaving sensing unobstructed.

Touch Norms and Moral Patiency. Soft plush exteriors increased willingness for social touch and reluctance to mistreat devices (RFPM: *materiality/texture* → warmth/moral patiency → touch norms). However, we caution against "trust-washing" via cute apparel for high-stakes roles (e.g. security, medical triage), reinforcing the need for domain-appropriate guardrails (G4–G5). *"It reminds me of a psychology experiment [33]. They gave a baby monkey two options: one that looks like its mother but is made of cold steel, and a warm blanket that doesn't look like her at all. It picked the blanket. I think it's the same with robots"* – Eren Chen, Booster Robotics [9].

Capability Scripts and Expectancy Violations. A central mechanism in RFPM is that dress invites observers to project a *capability script* (what the robot should do, how reliable it should be, and what norms govern it). When attire implies a script the robot does not execute (e.g. medical aesthetics without triage policy, "security" cues without robust safety behavior), expectancy violations can degrade trust and produce harm. This motivates capability-honest styling and disclosure (G5), and treating attire as part of the robot's behavior contract rather than a cosmetic layer.

Normativity and Stereotype Risk. Because dress encodes cultural scripts, robot fashion can inadvertently reproduce gendered or normative expectations (e.g. "professional" tailoring reading as masculinized competence, or costumes reinforcing caricatures). For a foundational research agenda, this is not peripheral: normativity shapes who receives authority, who feels invited to interact, and which scripts become default. Design guardrails include avoiding credentialed aesthetics without matching policies (G5), auditing symbols across communities (G7), and preferring capability-honest cues over culturally loaded tropes. *"Adding human clothing, such as skirts, wigs, pasties (see Fig. 1a), causes shock and even disgust from some people. There's a line where costume stops being fun and starts feeling wrong"* – Nima Zeighami, REK [46].

Situated Exemplar (fight club; costume as persona). Fight-club teams frequently treat attire as *costume*: a fast, spectator-readable persona layer that produces character expectations and narratives (e.g. "hero," "villain," "cowboy," "mascot") while still needing to preserve motion and safety. *"It creates a reason for people to root for characters – especially if there are predetermined rivalries. At Stanford versus Berkeley, the Berkeley robot had bear ears. People were shouting 'let's go Stanford!' and 'drop those ears!' The outfit gave them something to fight for"* – Vitaly Bulatov, Ultimate Fighting Bots [52].

From a fashion-psych lens, these costumes operate as *impression management* and *group identity signals*, often shifting warmth/competence readings depending on silhouette, materials (soft vs. hard), and iconography. Fight-club attire hints at *resistance practices*: teams dress robots in deliberately transgressive or ironic costumes that subvert expected readings – a phenomenon worth studying as robot fashion norms stabilize and invite pushback.

Situated Exemplar (industry; uniform as legibility and brand). In robotics-company demos and deployments, attire more often functions as *uniform*: an institutional signal intended to communicate affiliation, purpose, and safety conventions to the public (Fig. 1). *"The wraps, we wanted Sui colors (see Fig. 3b). It was a branding opportunity. People started calling them 'the Sui dogs.' It wasn't just 'robot dogs', it was 'these are the robot dogs that Sui has.' Otherwise, it would just be 'oh, they got Unitrees.' The outfit gave them identity"* – John Naulty, Mysten Labs [45].

4.4 Functional and Aesthetic Design Considerations

Robot garments are *interfaces*, not mere decoration: they mediate protection, thermals, sensing, and expression [26]. Table 3 operationalizes these constraints as guidelines G1–G3 (kinematics, thermals, serviceability) and G8 (lifecycle). In practice, protection is often also *aesthetic* (shells communicating robustness or play), while modularity (snaps, magnets, zip-off panels) is a pragmatic driver of style: outfits must be removed for charging, cleaning, or inspection. These serviceability requirements imply invisible human labor infrastructure that will become salient. *"You don't want super loose stuff getting caught in the actuators or joints. If you're doing a backflip and step on your clothes, you might not be doing a backflip at the end"* – John Naulty, Mysten Labs [45]. Interactive affordances (e-textiles, thermochromic threads, light-emitting garments) can improve state legibility but introduce maintenance and failure pathways (G6) [19, 39, 50]. Color, silhouette, and cultural references require sensitivity (G7); visibility choices (e-ink, high-vis palettes) avoid deception in public spaces [26].

4.5 Signal Integrity and Accountable Wardrobe Autonomy

Because attire can shift perceived authority and trust (RFPM: *uniform/livery* and *silhouette/tailoring* → authority heuristics → compliance), high-stakes contexts motivate *signal integrity*: the ability to bound, audit, and disclose what an outfit is meant to imply. This extends "ownership" beyond fabric to questions of authorship, authorization, and accountability – and positions wardrobe as part of a robot's public-facing safety contract (G4 – G5).

Today: Human Proxy Ownership. Wardrobes are chosen and owned by operators, brands, or venues; decals and accessories personalize platforms. This raises unresolved tensions: corporate branding may prioritize fleet recognition over user preferences; operators may resist institutional dress codes; and public policy is only beginning to consider robot appearance as a regulated domain. Robot fashion as described also requires expertise (patternmaking, e-textiles) and resources that are unevenly distributed; without democratizing tools and practices, fashionable robots may become another axis of inequality – available to well-resourced operators while others deploy "naked" or ill-fitting machines.

Digital Fashion & Provenance. Digital records can support traceability and lifecycle management (e.g. garment passports, fleet inventory, care and

Table 3: Morphology-aware garment guidelines. Operationalize RFPM mechanism families (Table 1) under morphology constraints and PIDI functions (Table 2). Includes rationale, example, and typical failure mode observed or reported in practice.

ID	Guideline (what / why)	Example implementation	Failure mode if ignored
G1	**Preserve kinematics & sensing.** Never occlude joints or sensor FOV; pattern for robot, not human body. *(RFPM: feasibility gate)*	Shoulder/elbow gussets; lapel camera window; quadruped cuffs clearing servos.	Restricted ROM; servo stall; camera occlusion; blind spots.
G2	**Plan thermal paths.** Venting and heat paths as first-order constraints. *(RFPM: feasibility gate)*	Back-yoke mesh + spacers; drone shell with prop-wash channels.	Heat soak; throttling; discoloration; touch discomfort.
G3	**Design for don / doff & service.** Fast attachment, laundering, and repair. *(RFPM: lifecycle reliability)*	Spine magnet rails; snap-on panels; zip-off sleeves; interior cable access.	Maintenance friction; dirty garments; operators remove clothing permanently.
G4	**Ensure signaling legibility & safety overlays.** Keep safety/ID layers readable independent of fashion. *(RFPM: authority & trust calibration)*	High-vis piping + ID badge separate from decorative layer.	"Trust-washing"; deception; public confusion; policy violations.
G5	**Align attire with behavior scripts.** Don't imply roles / capabilities you don't deliver. *(RFPM: capability scripts / expectancy)*	Avoid medical scrubs unless triage / clinic workflows are implemented.	Expectancy violations; over-trust; complaints or policy breaches.
G6	**Integrate interactive affordances judiciously.** Use e-textiles for touch/lighting/status where helpful. *(RFPM: feedback loops / state legibility)*	Capacitive touch strips; collar status LEDs.	Power-draining gimmicks; fragile "bling" that fails in service.
G7	**Respect culture & accessibility.** Avoid appropriation; support sensory needs and tactile landmarks. *(RFPM: social identity / normativity)*	Neutral civic palettes; tactile seams for wayfinding; non-irritant fabrics.	Exclusion; offense; sensory overload; reduced approachability.
G8	**Design for lifecycle.** Prefer modular, repairable, recyclable constructions. *(RFPM: durability supports trust)*	Replaceable sleeves; standardized fasteners; take-back plans.	Textile e-waste; short lifespan; poor fleet hygiene.

safety classifications). Tokenized or AR-only "skins" are a speculative extension of the same governance problem: how to manage identity continuity and authorship when appearance is changed rapidly and at scale.

Agency and Selection. Policy-bounded autonomy could allow robots to select among safe/appropriate looks (learning from interaction outcomes). Centralized "style updates" for fleets mirror software updates; balance uniformity with contextual fit.

Continuity and Identity. Attire can carry identity across embodiments (signature color/emblem), serving as a continuity marker for long-lived agents. *"If they were not unique, it would not have been as good. It's the same reason generic drugs are cheaper than branded ones, it's just a name. But, that uniqueness is the identity. Other people don't know what group you're in, but they know you're not vanilla"* – John Naulty, Mysten Labs [45].

Sketch: Accountable Wardrobe Autonomy. A garment-passport architecture (see supplementary materials) could link physical tags to passport services and audit logs so wardrobe policies can be enforced and inspected. We include this as an *implication mechanism* – not a finalized system – for bounding appearance autonomy with accountability and disclosure (RFPM: robot-facing codes → coordination benefits, also human auditability risks).

4.6 Speculative Scenarios: Robot-to-Robot Fashion Dynamics

We articulate near-future scenarios that probe robot-facing uses of fashion as a communication medium. In RFPM terms, these scenarios stress-test the "robot-facing codes" pathway (Table 1) and make explicit the trade-off between coordination benefits and human interpretability. *"We're already working on robots recognizing each other by what they wear. In multi-robot scenarios, visual markers on the outfit become part of the coordination layer. Not just decoration"* – Nima Zeighami, REK [46].

Compliments & Etiquette. Brief exchanges where classifiers trigger polite "nice livery" messages – a low-stakes social lubricant that can shape public encounter dynamics. **Social-grooming Analogs.** Robots adjust peers' badges/capes to maintain visibility or sensor access, signaling cooperation and care. **After-hours Fashion Play.** Autonomous agents project patterns for one another (LED/IR), exploring novelty and peer recognition as internal rewards. **Machine-vision-only Fashion.** IR/UV/fiducial weaves readable by robots but less salient to humans can enable coordination without cluttering the human visual field. **Robot Fashion Autonomy.** A more radical extension asks what fashion means when robots have full agency – including economic agency via permissionless financial systems (e.g. cryptocurrency without ID verification) [13, 14]. If robots can own themselves and their resources, they become fashion *subjects*, not just wearers: capable of purchasing, choosing, and expressing identity through dress. This shifts RFPM from a human-perception model to a multi-agent model where robots dress for themselves and each other, stress-testing the anthropocentric assumption that robot fashion is ultimately *for* human benefit.

These fictions surface questions around transparency, etiquette, group norms, and bias (e.g. exclusion via outdated codes), while motivating design directions for optical/tactile signaling garments. They complement empirical HRI by framing testable hypotheses for attire-mediated interaction.

5 Design Gallery: Morphology Case Studies

We present a brief gallery to materialize robot fashion across morphologies and to ground the guidelines in Table 3. Figure 3 showcases our morphology-diverse capsule collection spanning tabletop social robots, quadrupeds, humanoids, and plushie exteriors.

Tabletop Social Robot Head (Fig. 3a): minimal accessories re-cast persona while preserving sensing via seam placement and elastic attachments [16]. **Quadruped** (Fig. 3b): soft props vs. service livery elicit distinct pet/worker schemas while respecting clearance constraints. **Humanoid** (Fig. 3c): one wears a 3D knitted sweater shaped to sit below the chest screen, demonstrating robot-native fabrication; the other wears a varsity jacket and jeans, showing custom-fitted human-garment adaptation. **Plushie Exteriors** (Fig. 3d): soft covers and modular shells transform character while maintaining tactile affordances.

6 Speculative Outlook

Looking ahead a decade beyond, we expect robot fashion to become ordinary, infrastructural, and culturally expressive. We frame these projections through RFPM (Fig. 2): as attire cues become standardized and proliferate in the wild (Fig. 1), their psychological mechanisms (Table 1) will stabilize into norms, conventions, and points of contestation. The scenarios below extend our themes (Sec. 4) to articulate opportunities, risks, and an RFPM-driven agenda for design and evaluation.

Normalization and Infrastructure. Dressing robots becomes unremarkable: domestic agents ship with wardrobes; public-facing robots adopt uniforms and high-visibility layers. "Robo-tailors" emerge, bridging pattern-making, kinematics, and compliance; standardized attachment conventions (G1–G3, G8) become as expected as charging standards. Adaptive garments shift fit and signaling with task state (a "wearable check-engine light"), expanding the RFPM *interactivity* pathway into a state-readable channel. *"Clothes are just one part of it, accessories will be huge. Speakers that look like*

Figure 3: Morphology-diverse capsule collection: (a) KAMI [16] tabletop social robot head with accessories, (b) Unitree Go2 quadruped with playful vs. service attire, (c) Pepper humanoid wearing 3D knitted sweater, and Unitree G1 humanoid wearing varsity jacket and jeans, (d) PlaiPin [54] pocket companion robot with interchangeable plushie skins. Details in Sec. 5.

mouths, antennas shaped like ears. The line between functional hardware and aesthetic expression is going to blur" – Nima Zeighami, REK [46].

Culture, Agency, and Inequality. Owners co-style companions; regional styles and subcultures emerge; within policy bounds, agents learn to select outfits that improve cooperation ("learning to dress"). Yet aesthetic divides may mirror socioeconomic ones, shaping service expectations. RFPM predicts *silhouette*, *materiality*, and *iconography* cues will function as class signals and require cross-cultural auditing (G7). **Governance and Blended Realities.** Standards require persistent identification regardless of outfit; attire must not imply credentials the robot lacks (G4–G5). AR overlays and "skin packs" allow rapid appearance changes, extending signal-integrity concerns to appearance-at-scale: when looks can be swapped instantly, auditability and bounded wardrobe policies become central (cf. Sec. 4.5). **RFPM-derived Hypotheses (testable agenda).** RFPM generates falsifiable questions rather than declare universal effect sizes. We highlight a set of testable hypotheses that follow Table 1:

- **H1 (Uniform → Authority).** Uniform/livery cues increase compliance, but also increase over-trust under capability mismatch; disclosure layers reduce the mismatch effect (G4–G5).
- **H2 (Texture → Touch).** Plush/soft materiality increases touch willingness and moral patiency, moderated by morphology (stronger for animaloids) and domain stakes.
- **H3 (Silhouette → Expectancy).** Tailoring/silhouette cues increase perceived competence and tighten expectancy; errors produce stronger negative reactions under "high competence" looks.
- **H4 (Iconography → Identity).** Affiliation and iconography cues increase bonding and naming, but also increase exclusion and bias under insider-coded styles; participatory auditing reduces harms (G7).
- **H5 (Robot-facing Codes → Coordination).** Machine legible attire improves robot-robot recognition/coordination, but decreases human auditability unless paired with human-legible overlays (G4).

Critical Tensions. We acknowledge tensions this paper cannot resolve. Is robot fashion inherently manipulative? If clothing systematically shifts trust and compliance, can any styling be truly "capability-honest," or is it designed influence by another name? Who benefits when robots are dressed? Our guidelines serve designers and operators, but the psychological effects land on humans who did not choose the outfit. What does it mean that robot fashion emerges at spectacle events, defaulting to off-the-shelf human clothing rather than robot-native designs? And who will do the invisible labor of dressing, maintaining, and laundering robot wardrobes? These questions deserve sustained engagement beyond design guardrails. Robot clothing is not ornament: it is a manipulable interface for safety, meaning, and belonging. RFPM offers a foundation for designing it with intention rather than improvisation.

7 Conclusion

We have presented *robot fashion* as a legitimate and timely domain at the intersection of HRI, social psychology, fashion studies, and materials engineering. Our central theoretical contribution is the **Robot Fashion Psychology Model (RFPM)** (Fig. 2), which formalizes robot attire as a psychological technology: *attire cue families* recruit predictable *mechanisms* (Table 1) that shape HRI outcomes – moderated by morphology, context and stakes, robot behavior scripts, and cultural norms. In doing so, we move robot fashion beyond "aesthetics" into a researchable design variable with testable predictions and clear ethical pressure points.

We operationalized RFPM through a functional taxonomy (PIDI; Table 2), garment guidelines (G1–G8; Table 3), and speculative scenarios that stress-test robot–robot fashion dynamics and signal-integrity challenges.

Implications. Treating appearance as *dynamic* rather than fixed expands the HRI toolbox. RFPM highlights why small attire cues can have outsized effects: they activate role schemas, competence/warmth inferences, identity signals, and capability scripts that shape how people approach, touch, comply with, or mistrust robots. Designers can use this to reduce ambiguity and support legibility, but must also treat wardrobe as part of a robot's public-facing behavior contract (G4–G5). We caution against "trust-washing," cultural insensitivity, and deceptive authority cues; transparency requirements (e.g. ID and capability disclosures independent of outfit) and accessibility practices should accompany any wardrobe strategy (G4, G7).

Future Work. Immediate threads include preregistered, cross-cultural studies testing RFPM pathways (recognizing this paper's Western perspective), dressing-aware hardware standards (G1–G3, G8), and governance mechanisms for accountable wardrobe autonomy. Sec. 6 details hypotheses (H1–H5) and risk mitigations. Longer term, robot-facing fashion may require new mechanism families beyond Table 1: *robot identity construction*, *robot peer perception*, and *robot fashion norms* – speculative extensions that future work should revisit as robot autonomy matures, inviting collaboration across HRI, fashion, materials, and ethics communities.

In short, as robots populate homes, streets, and skies, their clothes (fabric, shells, light, or digital skins) will shape how humans read robots and how robots coordinate. We offer RFPM as a foundation and invitation: to study, standardize, and design robot fashion thoughtfully so the social presence it creates serves safety, meaning, and belonging rather than spectacle alone.

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References

[1] Hajo Adam and Adam D. Galinsky. 2012. Enclothed Cognition. *Journal of Experimental Social Psychology* 48, 4 (2012), 918–925. doi:10.1016/j.jesp.2012.02.008

[2] Gabriela Ambás and Paula Gárgoles. 2025. Fashioning Society: An Exploration of Collective Behaviour and Cultural Change Through Symbolic Interactionism. *Filosofija. Sociologija* 36, 1 (2025), 50–59. doi:10.6001/fil-soc.2025.36.1.6

[3] Aayushi Badhwar, Saniyat Islam, and Caroline Swee Lin Tan. 2023. Exploring the Potential of Blockchain Technology within the Fashion and Textile Supply Chain with a Focus on Traceability, Transparency, and Product Authenticity: A

Systematic Review. *Frontiers in Blockchain* 6 (2023), 1044723. doi:10.3389/fbloc.2023.1044723

[4] Roland Barthes. 1983. *The Fashion System*. University of California Press, Los Angeles. Translated by Matthew Ward and Richard Howard.

[5] Russell W. Belk. 1988. Possessions and the Extended Self. *Journal of Consumer Research* 15, 2 (1988), 139–168. doi:10.1086/209154

[6] Silvia Bellezza, Francesca Gino, and Anat Keinan. 2014. The Red Sneakers Effect: Inferring Status and Competence from Signals of Nonconformity. *Journal of Consumer Research* 41, 1 (2014), 35–54. doi:10.1086/674870

[7] Leonard Bickman. 1974. The Social Power of a Uniform. *Journal of Applied Social Psychology* 4, 1 (1974), 47–61. doi:10.1111/j.1559-1816.1974.tb02599.x

[8] Herbert Blumer. 1969. Fashion: From Class Differentiation to Collective Selection. *The Sociological Quarterly* 10, 3 (1969), 275–291. doi:10.1111/j.1533-8525.1969.tb01292.x

[9] Booster Robotics. 2025. Booster Robotics. https://www.booster.tech. Accessed November 2025.

[10] Juan José Bullón Pérez, Araceli Queiruga-Dios, Víctor Gayoso Martínez, and Ángel Martín del Rey. 2020. Traceability of Ready-to-Wear Clothing through Blockchain Technology. *Sustainability* 12, 18 (2020), 7491. doi:10.3390/su12187491

[11] Brad J. Bushman. 1984. Perceived Symbols of Authority and Their Influence on Compliance. *Journal of Applied Social Psychology* 14, 6 (1984), 501–508. doi:10.1111/j.1559-1816.1984.tb02255.x

[12] Brad J. Bushman. 1988. The Effects of Apparel on Compliance: A Field Experiment with a Female Authority Figure. *Personality and Social Psychology Bulletin* 14, 3 (1988), 459–467. doi:10.1177/0146167288143004

[13] Irvin Steve Cardenas and Jong Hoon Kim. 2018. Robot-Human Agreements and Financial Transactions Enabled by a Blockchain and Smart Contracts. In *Companion of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (Chicago, IL, USA) *(HRI '18)*. Association for Computing Machinery, New York, NY, USA, 337–338. doi:10.1145/3173386.3177818

[14] Irvin Steve Cardenas and Jong-Hoon Kim. 2020. Robonomics: The Study of Robot-Human Peer-to-Peer Financial Transactions and Agreements. In *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction* (Cambridge, United Kingdom) *(HRI '20)*. Association for Computing Machinery, New York, NY, USA, 8–15. doi:10.1145/3371382.3380735

[15] Irvin Steve Cardenas, Jong-Hoon Kim, Margarita Benitez, Kelsey A Vitullo, Michelle Park, Chanjuan Chen, and Linda Ohrn-McDaniel. 2019. Telesuit: design and implementation of an immersive user-centric telepresence control suit. In *Proceedings of the 2019 ACM International Symposium on Wearable Computers* (London, United Kingdom) *(ISWC '19)*. Association for Computing Machinery, New York, NY, USA, 261–266. doi:10.1145/3341163.3346936

[16] Irvin Steve Cardenas, John Edward Naulty, George Digkas, Kostas Kryptos Chalkias, and Jong-Hoon Kim. 2025. KAMI: An Open Source HRI Platform for Social Interaction and Embodied AI. In *Open-Source Hardware Workshop at the Conference on Robot Learning (CoRL 2025)*. Seoul, Korea. https://open-hardware-robots.github.io/CoRL2025/

[17] Robert B. Cialdini and Noah J. Goldstein. 2004. Social Influence: Compliance and Conformity. *Annual Review of Psychology* 55 (2004), 591–621. doi:10.1146/annurev.psych.55.090902.142015

[18] IEEE RAS Humanoids 2025 Organizing Committee. 2025. Robot Fashion Show 2025: Call for Participation. https://robotfashion.show/call-for-participation. 24th IEEE-RAS International Conference on Humanoid Robots (Humanoids '25), Seoul.

[19] Laura Devendorf, Joanne Lo, Noura Howell, Jung Lin Lee, Nan-Wei Gong, M. Emre Karagozler, Shiho Fukuhara, Ivan Poupyrev, Eric Paulos, and Kimiko Ryokai. 2016. "I don't Want to Wear a Screen": Probing Perceptions of and Possibilities for Dynamic Displays on Clothing. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) *(CHI '16)*. Association for Computing Machinery, New York, NY, USA, 6028–6039. doi:10.1145/2858036.2858192

[20] Joanne Entwistle. 2015. Bruno Latour: Actor-network theory and fashion. In *Thinking Through Fashion: A Guide to Key Theorists*, Agnès Rocamora and Anneke Smelik (Eds.). I.B. Tauris, London, 269–284.

[21] Joanne Entwistle. 2015. *The Fashioned Body: Fashion, Dress, and Social Theory*. Polity Press, Oxford.

[22] Susan T. Fiske, Amy J. C. Cuddy, Peter Glick, and Jun Xu. 2002. A Model of (Often Mixed) Stereotype Content: Competence and Warmth Respectively Follow from Perceived Status and Competition. *Journal of Personality and Social Psychology* 82, 6 (2002), 878–902. doi:10.1037/0022-3514.82.6.878

[23] Kate Fletcher. 2016. *Craft of Use: Post-Growth Fashion*. Routledge, London.

[24] Natalie Friedman, Awsaf Ahmed, and Wendy Ju. 2024. Skins, Clothes, Costumes, Bling: How and Why Experts Dress their Robots. In *Proceedings of the Halfway to the Future Symposium* (Santa Cruz, CA, USA) *(HttF '24)*. Association for Computing Machinery, New York, NY, USA, Article 34, 10 pages. doi:10.1145/3686169.3686187

[25] Natalie Friedman, Kari Love, Alexandra Bremers, A.J. Parry, Ray LC, Bolor Amgalan, Jen Liu, and Wendy Ju. 2021. Designing Functional Clothing for Human-robot Interaction. In *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction* (Boulder, CO, USA) *(HRI '21 Companion)*. Association for Computing Machinery, New York, NY, USA, 703–705. doi:10.1145/3434074.3444870

[26] Natalie Friedman, Kari Love, Ray LC, Jenny E. Sabin, Guy Hoffman, and Wendy Ju. 2021. What Robots Need From Clothing. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (DIS '21)*. ACM, New York, NY, USA, 1345–1355. doi:10.1145/3461778.3462045

[27] Sofia Gioumatzidou, Anna-Maria Valentza, Katrin Fischer, and Nikolaos Fachantidis. 2024. 'Do you trust me with your secret if I wear the clothes you made for me?' Kids knitting clothes for SARs and choosing one to share a secret wish. In *Proceedings of the 11th International Workshop on Social and Cognitive Interactions for Assistive Robotics (SCRITA 2024), HRI (CEUR Workshop Proceedings, Vol. 3906)*. CEUR-WS.org, Aachen, Germany, 8 pages.

[28] Erving Goffman. 1959. *The Presentation of Self in Everyday Life*. Doubleday, Scotland.

[29] Heather M. Gray, Kurt Gray, and Daniel M. Wegner. 2007. Dimensions of Mind Perception. *Science* 315, 5812 (2007), 619–619. doi:10.1126/science.1134475

[30] Kurt Gray, Adam Waytz, and Liane Young. 2012. The Moral Dyad: A Fundamental Template Unifying Moral Judgment. *Psychological Inquiry* 23, 2 (2012), 206–215. doi:10.1080/1047840X.2012.686247

[31] Kurt Gray and Daniel M. Wegner. 2009. Moral Typecasting: Divergent Perceptions of Moral Agents and Moral Patients. *Journal of Personality and Social Psychology* 96, 3 (2009), 505–520. doi:10.1037/a0013748

[32] Kurt Gray and Daniel M. Wegner. 2012. Feeling Robots and Human Zombies: Mind Perception and the Uncanny Valley. *Cognition* 125, 1 (2012), 125–130. doi:10.1016/j.cognition.2012.06.007

[33] Harry F. Harlow. 1958. The Nature of Love. *American Psychologist* 13, 12 (1958), 673–685. doi:10.1037/h0047884

[34] Ernst Harms. 1938. The Psychology of Clothes. *Amer. J. Sociology* 44, 2 (1938), 239–250. doi:10.1086/217972

[35] Neil Hester and Eric Hehman. 2023. Dress is a Fundamental Component of Person Perception. *Personality and Social Psychology Review* 27, 4 (2023), 414–433. doi:10.1177/10888683231157961

[36] Koen V. Hindriks, Marijn Hagenaar, and Anna Laura Huckelba. 2022. Effects of Robot Clothing on First Impressions, Gender, Human-Likeness, and Suitability of a Robot for Occupations. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. 428–435. doi:10.1109/RO-MAN53752.2022.9900771

[37] C. Blaine Horton Jr., Hajo Adam, and Adam D. Galinsky. 2025. Evaluating the Evidence for Enclothed Cognition: Z-Curve and Meta-Analyses. *Personality and Social Psychology Bulletin* 51, 2 (2025), 203–221. doi:10.1177/01461672231182478

[38] Yuniya Kawamura. 2005. *Fashion-ology: An Introduction to Fashion Studies*. Berg Publishers, Oxford.

[39] Huanhuan Liu, Yuyuan Shi, You Pan, Zhaohui Wang, and Bingjie Wang. 2025. Sensory interactive fibers and textiles. *npj Flexible Electronics* 9, 23 (2025), 12 pages. doi:10.1038/s41528-025-00398-4

[40] Carolyn Mair. 2024. *The Psychology of Fashion* (2 ed.). Routledge, London. doi:10.4324/9781032712550

[41] W. David Marx. 2022. *Status and Culture: How Our Desire for Social Rank Creates Taste, Identity, Art, Fashion, and Constant Change*. Viking, New York.

[42] Merriam-Webster. 2025. Garment. https://www.merriam-webster.com/dictionary/garment Accessed 5 November 2025.

[43] Stanley Milgram. 1963. Behavioral Study of Obedience. *Journal of Abnormal and Social Psychology* 67, 4 (1963), 371–378. doi:10.1037/h0040525

[44] Janice Catherine Miller. 2015. Sigmund Freud: More than a fetish: fashion and psychoanalysis. In *Thinking Through Fashion: A Guide to Key Theorists*, Agnès Rocamora and Anneke Smelik (Eds.). I.B. Tauris, London, 46–56.

[45] Mysten Labs. 2025. Mysten Labs. https://www.mystenlabs.com. Accessed November 2025.

[46] REK. 2025. REK – Robot Entertainment and Combat. https://rek.com. Accessed November 2025.

[47] Mary Ellen Roach-Higgins and Joanne B. Eicher. 1992. Dress and Identity. *Clothing and Textiles Research Journal* 10, 4 (1992), 1–8. doi:10.1177/0887302X9201000401

[48] Georg Simmel. 1991. The Problem of Style. *Theory, Culture and Society* 8, 3 (1991), 63–71. doi:10.1177/026327691008003004

[49] Georg Simmel. 1998. The Philosophy of Fashion. In *Simmel on Culture: Selected Writings*, David Frisby and Mike Featherstone (Eds.). Sage Publications, London, 188–189.

[50] Takashi Sugiyama and Masayoshi Kanoh. 2023. Investigating Emotional Expressivity in Robots Wearing Light-Emitting Clothing. In *Proceedings of the 2023 AHFE International Conference on Human Factors and Ergonomics in Healthcare and Medical Devices (AHFE 2023) and Affiliated Conferences*. AHFE International, USA, 8 pages. doi:10.54941/ahfe1004414 OpenAccess CMS Proceedings.

[51] Lou Taylor. 2002. *The Study of Dress History*. Manchester University Press, Manchester.

[52] Ultimate Fighting Bots. 2025. Ultimate Fighting Bots. https://ufb.gg. Accessed November 2025.

[53] Adam Waytz, Kurt Gray, Nicholas Epley, and Daniel M. Wegner. 2010. Causes and Consequences of Mind Perception. *Trends in Cognitive Sciences* 14, 8 (2010),

383–388. doi:10.1016/j.tics.2010.05.006

[54] Natalie Catherine Yeo and Irvin Steve Cardenas. 2026. PlaiPin: Designing Scalable, Safe, and Socially Acceptable Physical AI Companions Through Modularity. In *ACM/IEEE International Conference on Human-Robot Interaction (HRI '26), Industry Paper*. Edinburgh, Scotland, UK. Industry paper.

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